

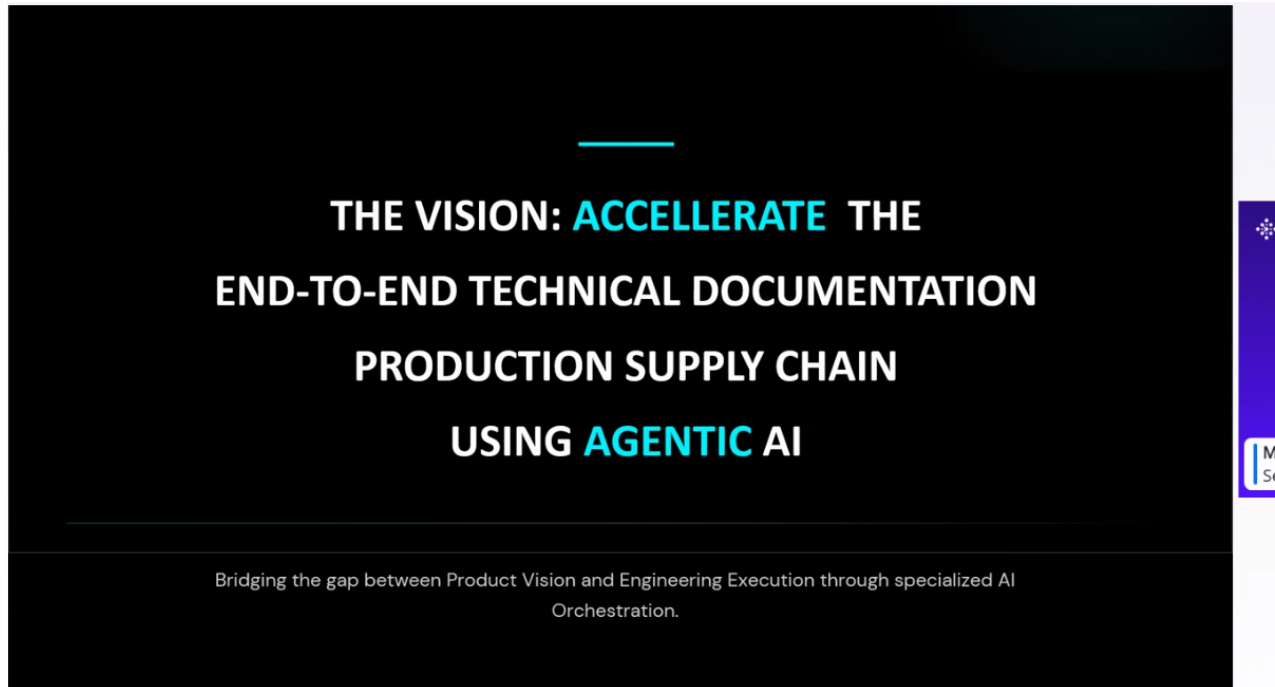
Digital Twin Knowledge Graph

Accelerating the Technical Documentation Supply Chain with Agentic AI

A knowledge graph practitioner's reading of the talk slides

October 08, 2026

1. The Vision

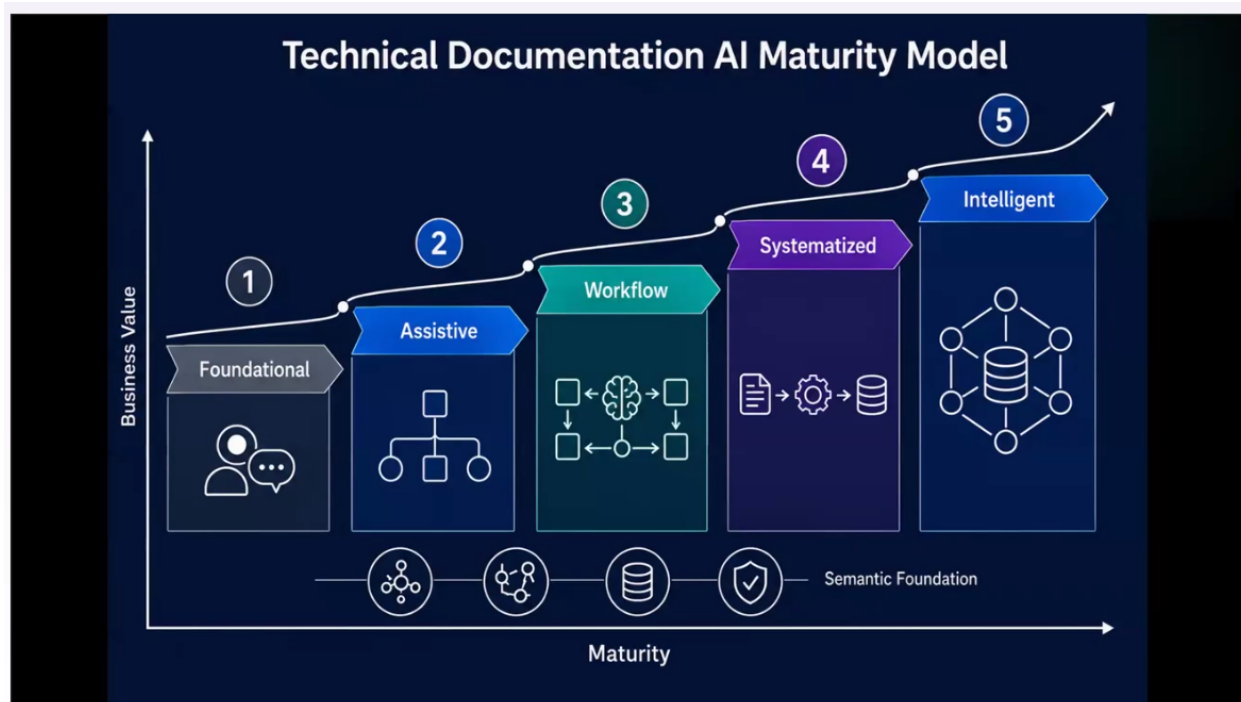


Slide: The Vision – accelerate the end-to-end technical documentation production supply chain using agentic AI

The talk frames technical documentation as a **supply chain**: product intent flows through engineering, authoring, review, translation and publishing. The stated aim is to bridge the gap between product vision and engineering execution through **specialized AI orchestration** – multiple agents each owning a stage – rather than a single general-purpose chatbot.

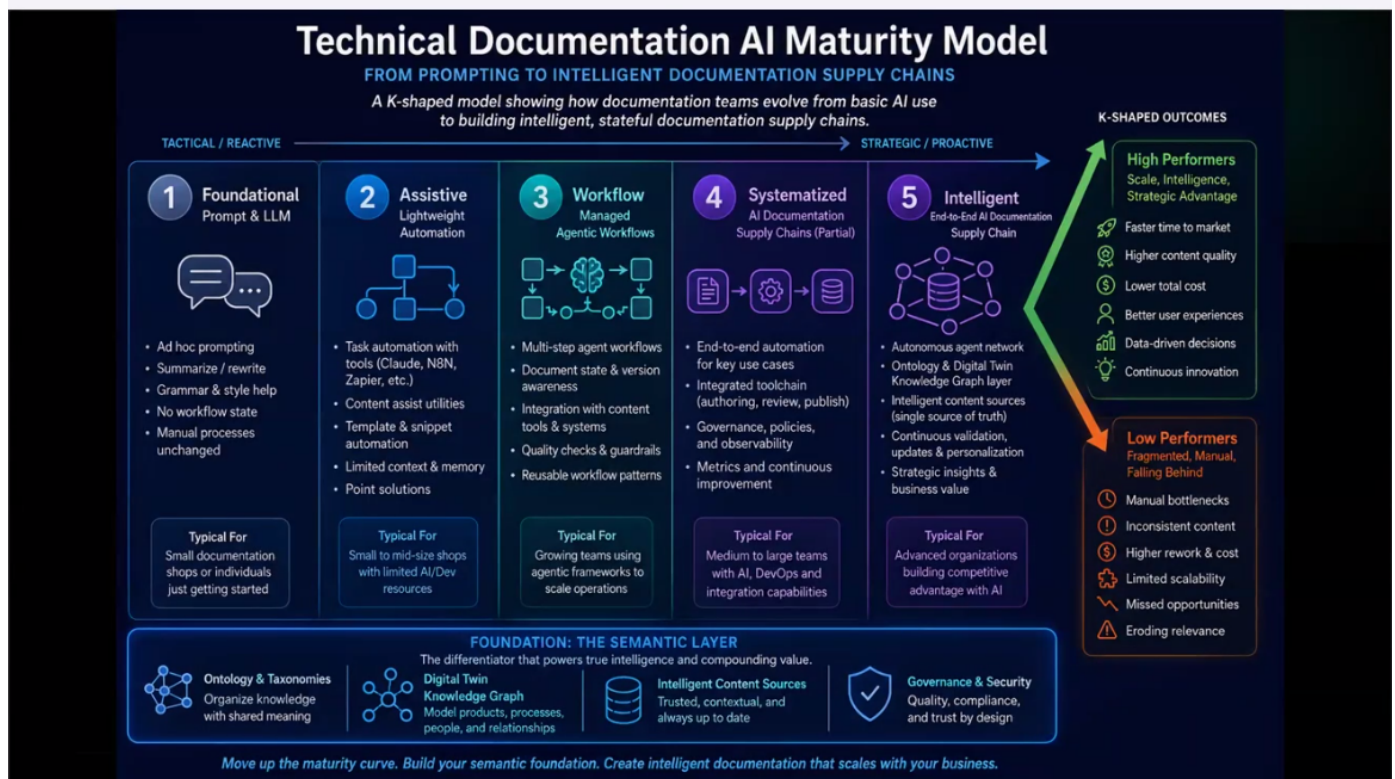
Expert read: the word 'orchestration' is the key. Agents only coordinate reliably when they share an explicit model of what content exists, how it is structured and what depends on what. That shared model is the knowledge graph introduced in later slides.

2. Technical Documentation AI Maturity Model



Slide: Maturity model – business value rises with maturity; a semantic foundation underpins every level

Five stages are plotted against business value: **Foundational** → **Assistive** → **Workflow** → **Systematized** → **Intelligent**. The base strip shows the **Semantic Foundation** (ontology/taxonomy, knowledge graph, data, governance) spanning all stages – value compounds only when that layer is present.



Slide: Detailed maturity model – 'From prompting to intelligent documentation supply chains' with K-shaped outcomes

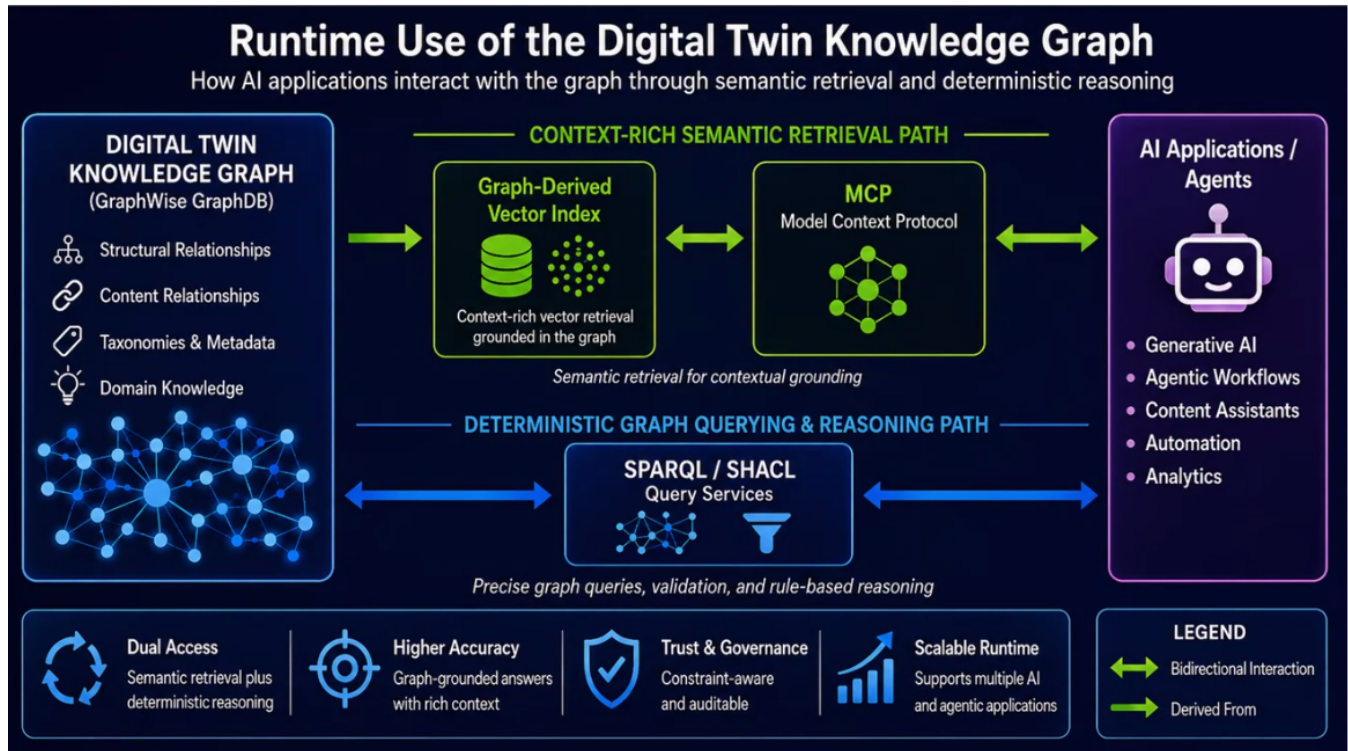
Maturity levels in detail

Level	Name	Characteristics	Typical for
1	Foundational – Prompt & LLM	Ad hoc prompting; summarize/rewrite; grammar and style help; no workflow state; manual processes unchanged	Small shops or individuals just starting
2	Assistive – Lightweight automation	Task automation with tools (Claude, n8n, Zapier); content-assist utilities; template/snippet automation; limited context and memory; point solutions	Small to mid-size shops with limited AI/dev resources
3	Workflow – Managed agentic workflows	Multi-step agent workflows; document state and version awareness; integration with content tools; quality checks and guardrails; reusable workflow patterns	Growing teams scaling with agentic frameworks
4	Systematized – AI documentation supply chains (partial)	End-to-end automation for key use cases; integrated toolchain (author, review, publish); governance, policies, observability; metrics and continuous improvement	Medium/large teams with AI, DevOps and integration capability
5	Intelligent – End-to-end AI supply chain	Autonomous agent network; ontology and digital-twin knowledge graph layer; intelligent content sources (single source of truth); continuous validation, updates, personalization; strategic insight	Advanced organizations building competitive advantage with AI

K-shaped outcomes. The model predicts divergence. **High performers** gain scale, intelligence and strategic advantage: faster time to market, higher content quality, lower total cost, better user experience, data-driven decisions and continuous innovation. **Low performers** stay fragmented and manual: bottlenecks, inconsistent content, rework and cost, limited scalability, missed opportunities and eroding relevance.

The semantic layer as differentiator. The slide names four pillars: **Ontology & taxonomies** (shared meaning), **Digital-twin knowledge graph** (products, processes, people, relationships), **Intelligent content sources** (trusted, contextual, always current) and **Governance & security** (quality, compliance, trust by design).

3. Runtime Use of the Digital Twin Knowledge Graph



Slide: How AI applications interact with the graph through semantic retrieval and deterministic reasoning

The graph (hosted in GraphDB) holds **structural relationships, content relationships, taxonomies and metadata, and domain knowledge**. AI applications and agents – generative AI, agentic workflows, content assistants, automation, analytics – reach it through **two complementary paths**:

- **Context-rich semantic retrieval path:** a graph-derived vector index (embeddings built from the enriched graph rather than isolated text chunks), exposed to agents via the **Model Context Protocol (MCP)**. Used for fuzzy, contextual grounding.
- **Deterministic querying and reasoning path:** **SPARQL** for precise graph queries and **SHACL** for validation and rule-based reasoning. Used where exact, repeatable, auditable answers are required.

Benefit	Meaning
Dual access	Semantic retrieval plus deterministic reasoning
Higher accuracy	Graph-grounded answers with rich context
Trust & governance	Constraint-aware and auditable
Scalable runtime	Serves multiple AI and agentic applications from one graph

Expert read: this is the practical form of hybrid AI – probabilistic retrieval to find candidates, symbolic reasoning to decide. The vector index is *derived from* the graph, so the two never drift apart.

4. Smashing 'The Wall'

SMASHING “THE WALL”

- ✓ Direct addressability to every affected DITA source topic
- ✓ Fast, deterministic analysis across the full content corpus
- ✓ Consistent results - same query returns the same affected topics
- ✓ Dramatically reduces LLM processing time and token consumption
- ✓ Only relevant topics and context are sent to the LLM
- ✓ SPARQL / SHACL provide precise, explainable, auditable reasoning
- ✓ Prevents unintended changes to unrelated parts of a topic
- ✓ Enables scalable, cost-effective content updating
- ✓ Graph relationships provide product, taxonomy, structural, and domain context

Plus...

The digital twins (graph and vector)



1. Are always current – never contains old chunks



2. Easily updated with an automated workflow daily (~ 30 min for 10,000 topics)

Fast. Accurate. Deterministic. Consistent. Explainable

Pssst: You can't do all of this with MD and Git

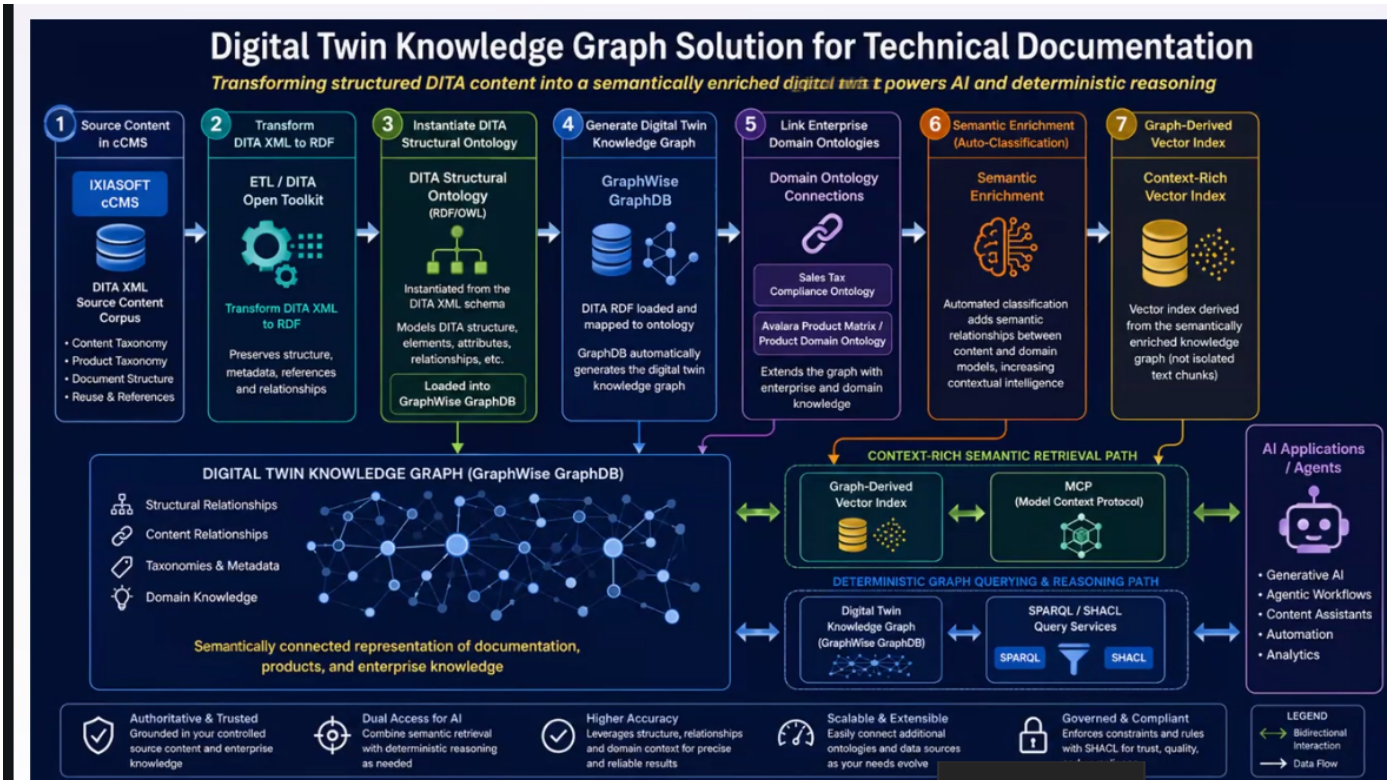
Slide: Why graph-based, deterministic analysis breaks through the documentation-update bottleneck

'The wall' is the point where an LLM-only approach to documentation change stops scaling: it cannot reliably know *which* topics a product change affects, so teams either send too much context or miss affected content. The slide lists what the graph provides instead:

- Direct addressability to every affected DITA source topic.
- Fast, deterministic analysis across the full content corpus; the same query returns the same affected topics.
- Only relevant topics and context are sent to the LLM, dramatically reducing processing time and token consumption.
- SPARQL/SHACL give precise, explainable, auditable reasoning and prevent unintended changes to unrelated parts of a topic.
- Enables scalable, cost-effective content updating; graph relationships supply product, taxonomy, structural and domain context.

The digital twins (graph and vector). (1) They are always current – they never contain stale chunks. (2) They are updated by an automated daily workflow, about **30 minutes for 10,000 topics**. The slide's closing line: *Fast. Accurate. Deterministic. Consistent. Explainable* – and, with a wink, 'you can't do all of this with MD and Git'.

5. The Solution Pipeline



Slide: Digital Twin Knowledge Graph solution – transforming structured DITA content into a semantically enriched digital twin

#	Step	What happens
1	Source content in CCMS	DITA XML corpus in a component content management system, with content and product taxonomies, document structure, and reuse/references
2	Transform DITA XML to RDF	ETL / DITA Open Toolkit converts XML to RDF, preserving structure, metadata, references and relationships
3	Instantiate DITA structural ontology	RDF/OWL ontology of DITA structure (elements, attributes, relationships) derived from the DITA schema and loaded into GraphDB
4	Generate digital twin knowledge graph	DITA RDF is loaded and mapped to the ontology; GraphDB generates the graph automatically
5	Link enterprise domain ontologies	Domain ontologies (e.g. sales-tax compliance, product matrix / product-domain ontology) extend the graph with enterprise context
6	Semantic enrichment (auto-classification)	Automated classification adds semantic relationships between content and domain models, increasing contextual intelligence
7	Graph-derived vector index	Context-rich vector index built from the enriched graph, not from isolated text chunks

The graph then serves AI applications and agents through the dual paths shown in section 3. Stated outcomes: **authoritative and trusted** (grounded in controlled source content), **dual access for AI**, **higher accuracy**, **scalable and extensible** (add ontologies and data sources as needs evolve) and **governed and compliant** (SHACL-enforced constraints and rules).

6. Key Takeaways

- **Move up the maturity curve by building the semantic foundation first** – it is the stated differentiator between levels, and the source of the K-shaped divergence.
- **Model the documentation itself.** Converting DITA to RDF under a structural ontology makes every topic, reuse relationship and reference addressable and queryable.
- **Combine, don't choose.** Vector retrieval for contextual grounding, SPARQL/SHACL for exact impact analysis and validation, both derived from one graph.
- **Send the LLM less.** Deterministic impact analysis narrows the corpus to affected topics, cutting tokens, latency and unintended edits.
- **Keep twins fresh.** Daily automated rebuild (~30 min / 10,000 topics) keeps graph and vector index current.
- **Extend with domain ontologies** (compliance, product matrix) so documentation reasoning reflects the enterprise, not just document structure.

Note: figures such as the 30-minute / 10,000-topic update time are quoted from the speaker's slides and were not independently verified.